ISM_NSP4_Lecture9_04.07.2026

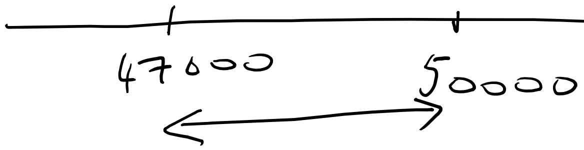
04 July 2026 08:11

$$\bar{X} = \frac{X_1 + X_2 + \dots X_{30}}{30} = 47000 \text{ Kms}$$

$$SD = \frac{1}{(n-1)} \sum_{i=1}^{n} (X_i - \bar{X})^2$$ Here $n = 30$

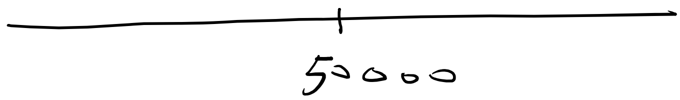
$$= \underline{5500 \text{ km}}$$

04 July 2026 08:34



You either have a rare sample or the original hypothesis of mean life of tyres being 50000kms is false

04 July 2026 08:41



sample standard deviation = 5500 km

standard deviation of sample means

$$= \frac{5500}{\sqrt{30}}$$

$$\bar{X} = \frac{X_1 + X_2 + \dots X_{30}}{30}$$

$$Var(\bar{X}) = \frac{1}{30^2} Var(X_1) + \frac{1}{30^2} Var(X_2) + \dots \frac{1}{30^2} Var(X_{30})$$

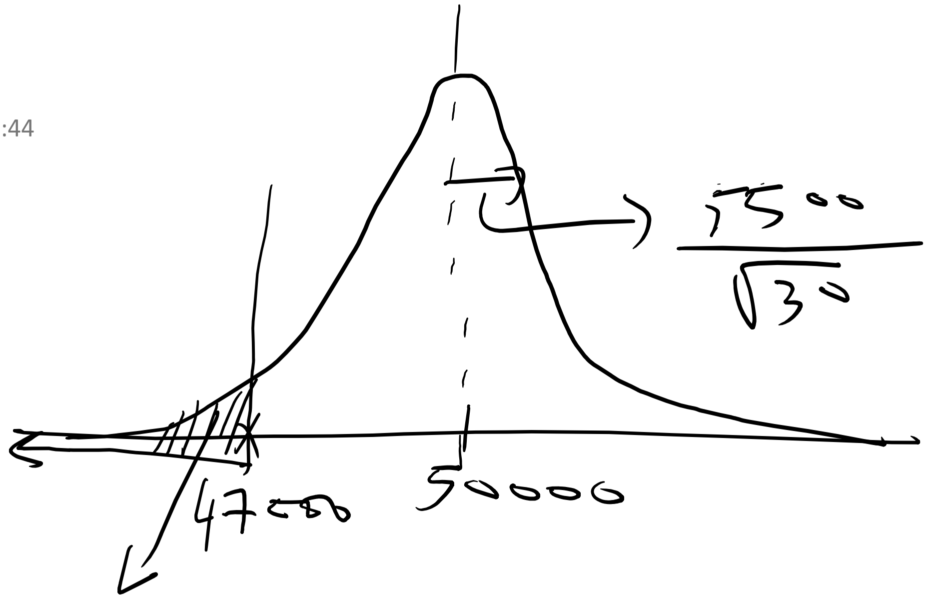
But $X_1, X_2, \dots X_{30}$ are independent identically drawn samples from the population

$$\therefore Var(X_1) = Var(X_2) = \dots Var(X_{30})$$

$$Var(\bar{X}) = \frac{30 \cdot Var(X_1)}{30^2} = \frac{Var(X_1)}{30}$$

$$\sqrt{Var(\bar{X})} = stddev(\bar{X}) = \frac{\sqrt{Var(X_1)}}{\sqrt{30}}$$

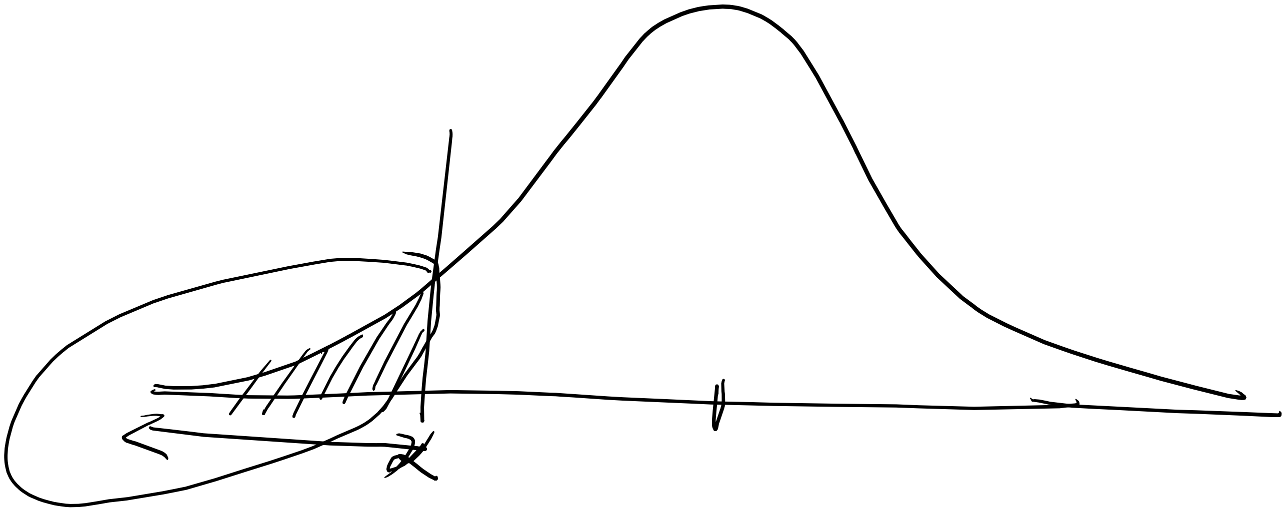
$$stddev(\bar{X}) = \frac{stddev(X_1)}{\sqrt{30}}$$



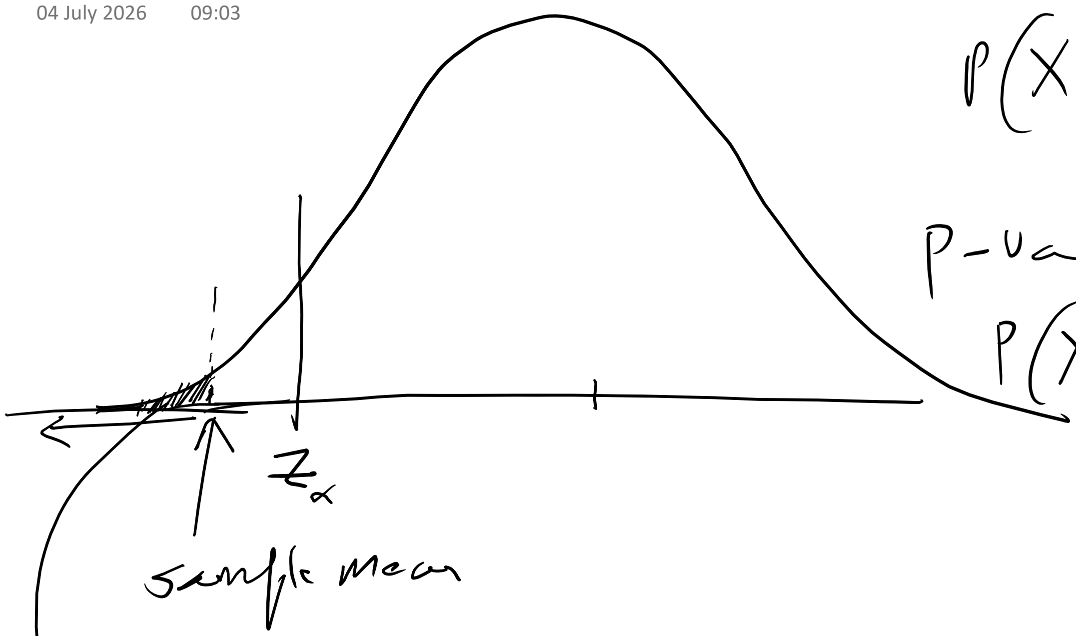
$$\frac{7500}{\sqrt{30}} = \sigma_{\bar{x}}$$

0.00013

04 July 2026 08:56



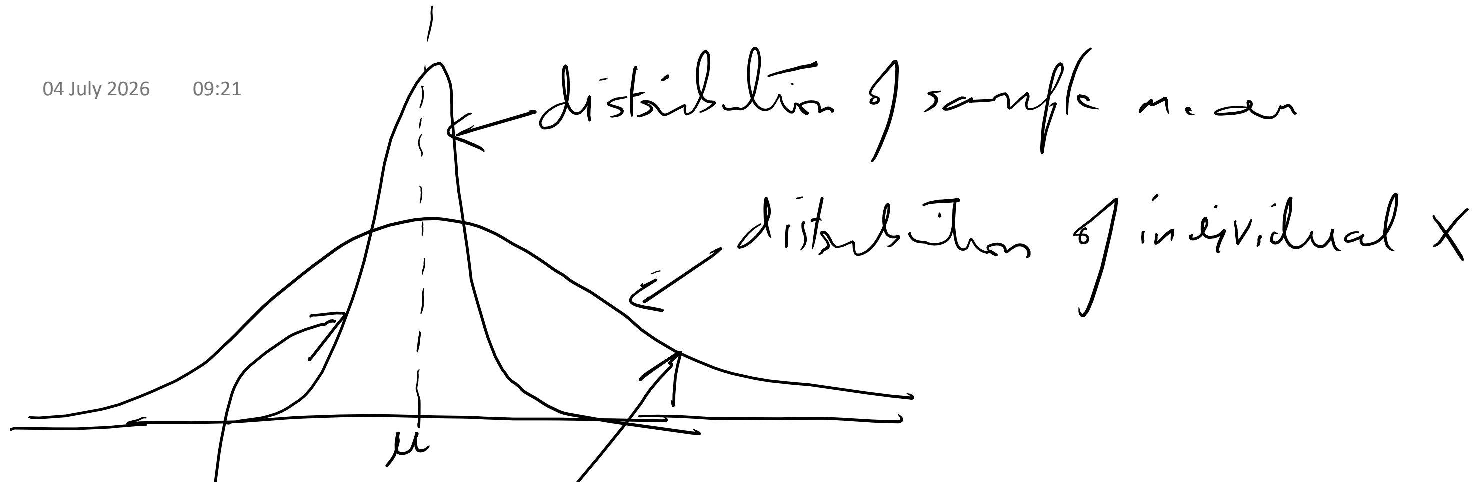
$$P(X \leq Z_\alpha) = \alpha$$



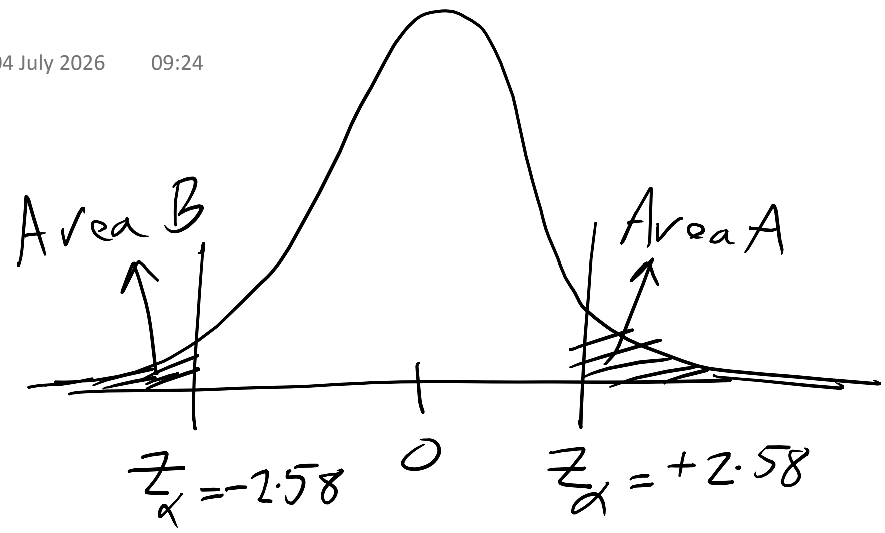
p-value

$$P(X \leq 5)$$

area here is p-value



$$\bar{X} = \frac{X_1 + X_2 + \dots X_n}{n}$$



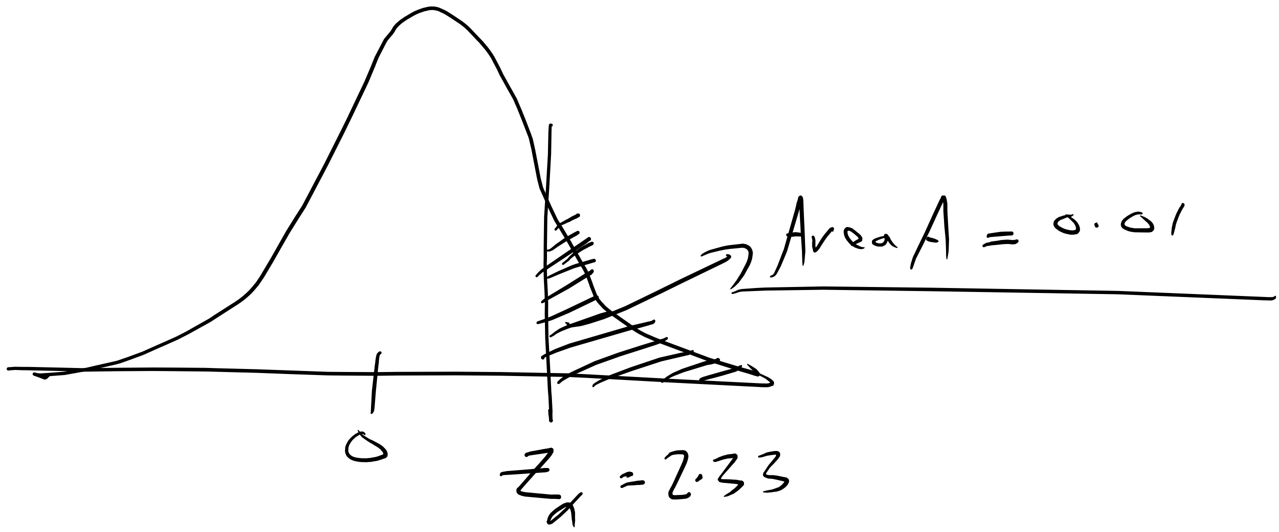
$|Z_\alpha| = 2.58$

level of significance

$\alpha = 0.01$

$$\text{Area } A = \text{Area } B = 0.005$$

$$\underline{\text{Area } A + \text{Area } B = 0.01}$$

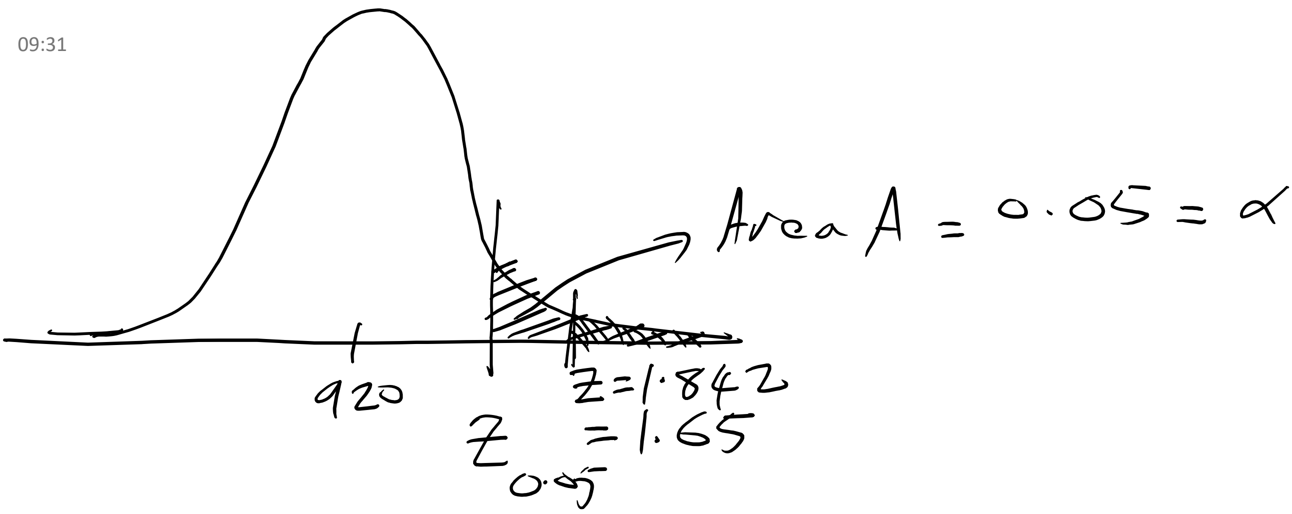

Area A = 0.01
0
$Z_\alpha = 2.33$

Recall that

$$s = \frac{1}{n-1} \sum_{i=1}^{n} \left(x_i - \bar{x}\right)^2$$

$$\bar{X} = \frac{X_1 + X_2 + \cdots X_n}{n}$$

std deviation of sample mean is the standard deviation of $\bar{X}$



$$\frac{925-920}{\frac{18}{\sqrt{44}}} = Z = 1.842$$

Area B is probability to the right of $Z = 1.842$

Area B < Area A

$$Z = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$Y = \frac{X_1^A + X_2^A + \cdots X_{n_1}^A}{n_1} - \frac{X_1^B + X_2^B + \cdots X_{n_2}^B}{n_2}$$

Each X_i^A and X_i^B are normal random varid and they are independent of all the other variables

$$Var(Y) = \frac{1}{n_1^2} n_1 Var(X_1^A) + \frac{1}{n_2^2} \cdot n_2 \cdot Var(X_1^B)$$

because $X_1^A, X_2^A \ldots X_{n_1}^A$ are all independent identically distributed random variables

$$Var(Y) = \frac{Var(X_1^A)}{n_1} + \frac{Var(X_1^B)}{n_2} = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$$

$$\sigma_p = \sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}} = \sigma_p$$

[drawing a sample of size n from N population items without replacement]

sampling with replacement

finite population correction factor

$$\bar{X} = \frac{X_1 + X_2 + \dots X_n}{n}$$

[in case of sampling with replacement]

Each X_i is a binary variable (0 or 1) and is independent of any other X_i

$P(X_i = 1) = p$ $\qquad$ $P(X_i = 0) = 1 - p$

$$Var(X_i) = E(X_i^2) - E^2(X_i)$$

$$E(X_i) = P(X_i = 1)\,1 + P(X_i = 0)\,0$$

$$= p$$

$$E(X_i^2) = P(X_i = 1)\,1^2 + P(X_i = 0)\,0^2$$

$$E(X_i^2) = P(X_i = 1)\,1 + [illegible]$$

$$= p$$

$$Var(X_i) = p - p^2 = p(1-p)$$

$$E(X_i) = p$$

$$Var(\bar{X}) = Var\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right)$$

$$= \frac{1}{n^2}\, n\, Var(X_1) = \frac{Var(X_1)}{n}$$

$$\therefore Var\left(\frac{X_1 + X_2 + \dots X_n}{n}\right) = \frac{p(1-p)}{n} = \frac{pq}{n}$$

$$\text{As } N \to \infty \quad \sqrt{\frac{N-n}{N-1}} \to 1$$